

KnowSelf

Agentic Knowledgeable Self-Awareness for LLM-Based Agents

Teaching agents to autonomously decide when to act, reflect, or seek knowledge

arxiv.org/abs/2504.03553

The Problem: Indiscriminate "Flood Irrigation"

1

Trajectory Imitation

ReAct: blindly imitates gold trajectories without understanding when help is needed

2

Trial-and-Error

Reflexion: always reflects even when the agent already knows the correct action

3

Knowledge-Augmented

KnowAgent, WKM: injects domain knowledge at every step — 100% usage regardless of need

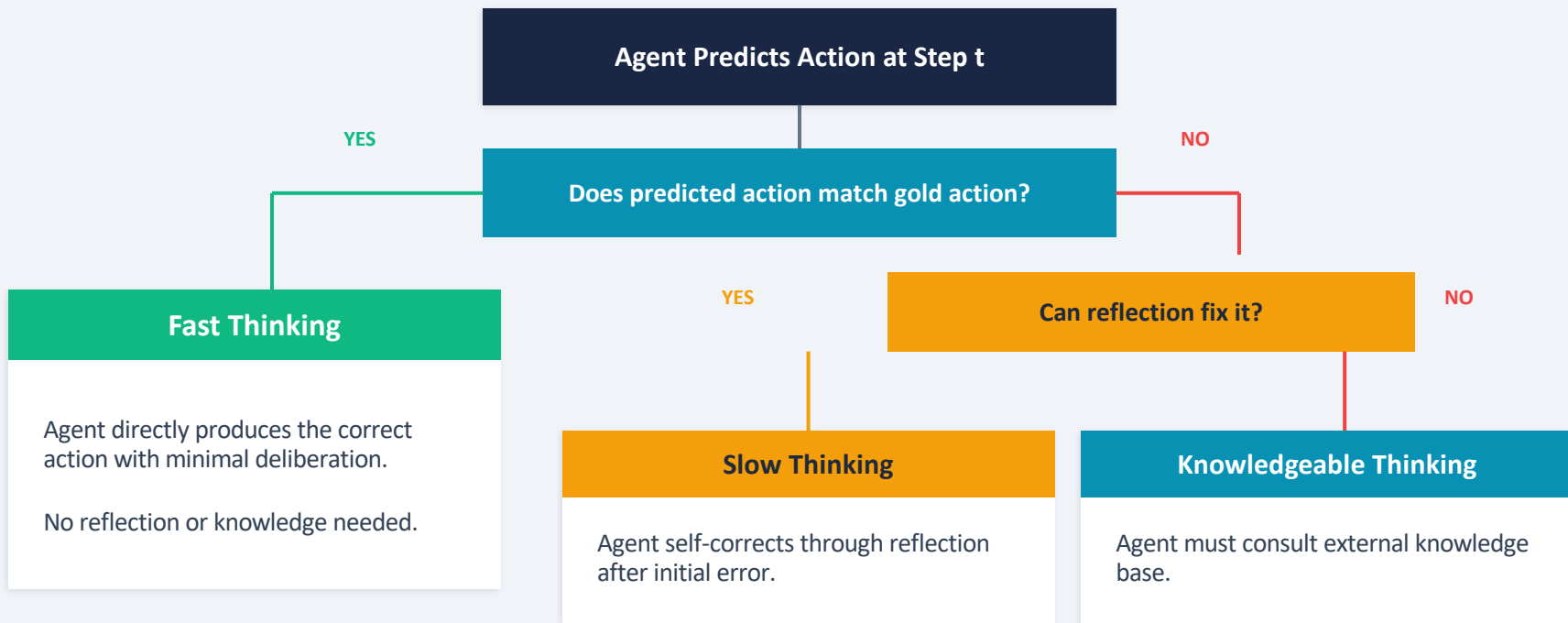
Human Approach

Humans possess situational self-awareness: the metacognitive ability to assess whether they can handle a situation directly, need to pause and reflect, or must seek external help.

KnowSelf brings this capability to LLM agents.

Result: fragile planning, pattern collapse, and unnecessarily high inference costs

Three Situational Thinking Modes



KnowSelf Framework Pipeline

1

Data Construction

- Agent self-explores environments
- Heuristic criterion classifies each step
- Special tokens inserted for situation type

2

Two-Stage Training

- Stage A: Supervised Fine-Tuning (SFT)
- Stage B: Reward-based Policy Optimization (RPO)
- Refines special token generation

3

Self-Aware Inference

- Agent generates special tokens at each step
- Tokens trigger: action / reflection / retrieval
- Knowledge retrieved only when needed

Lightweight and data-efficient: heuristic labeling on self-explored trajectories — no expensive human annotation

KnowSelf vs. Strong Baselines: Qualitative Examples

Task: "Put a hot mug in cabinet"

OpenAI-O1

Incorrectly tries to take an egg from the microwave after opening it. Fails to identify the mug heating step.

KnowSelf (Llama-8B)

Uses Reflection token to realize it should heat the mug using the open microwave. Correct action follows.

Task: "Put a saltshaker in drawer"

DeepSeek-R1

Reflects that drawer is open and tries to close it first — action fails. Gets stuck in loop.

KnowSelf (Llama-8B)

Applies Knowledge token: place the object without removing unrelated items. Succeeds.

KnowSelf selects the right cognitive mode — reflection where reflection helps, knowledge where knowledge is needed

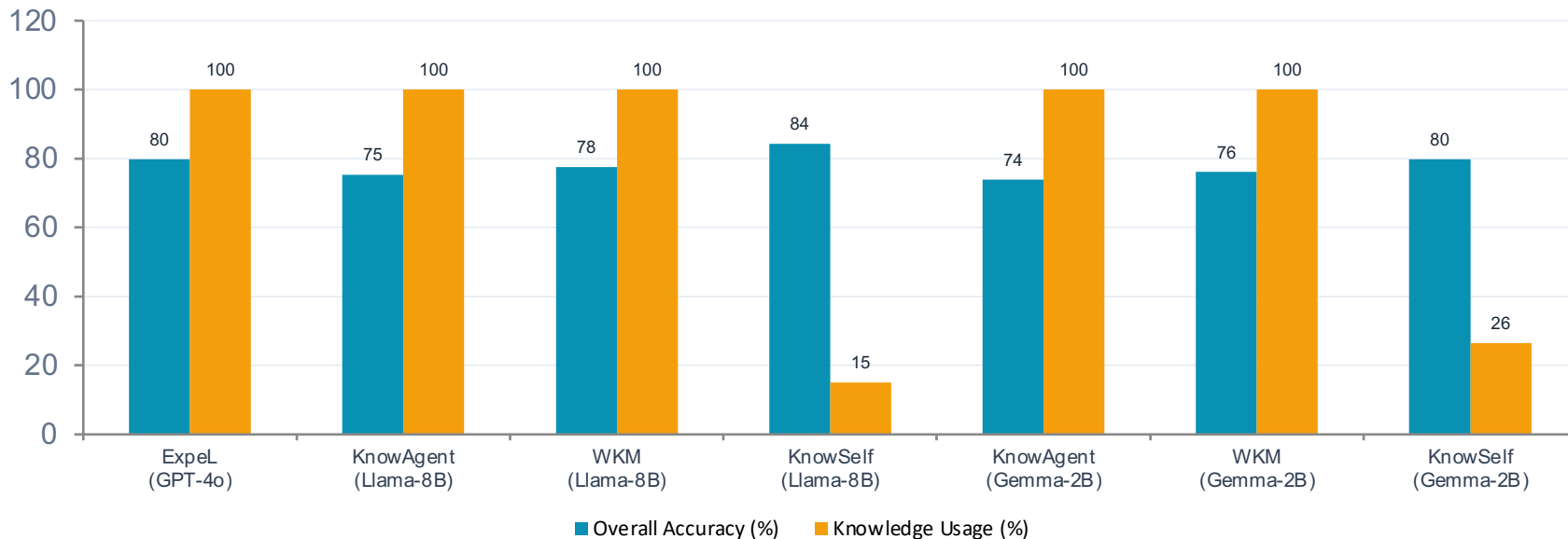
ALFWorld Results: 84.33% with Only 15% Knowledge

Backbone	Method	Know%	Put	Clean	Heat	Cool	Examine	Put Two	All
GPT-4o	ReAct	0%	83.33	74.19	69.57	85.71	77.78	64.71	76.12
GPT-4o	Reflexion	0%	100.0	87.10	73.91	90.48	83.33	70.59	85.07
GPT-4o	ExpeL	100%	95.83	83.87	69.57	80.95	88.89	52.94	79.85
Llama-8B	ReAct	0%	33.33	3.23	0.00	57.14	66.67	23.53	27.61
Llama-8B	Reflexion	0%	62.96	22.73	5.88	64.29	86.36	50.00	51.49
Llama-8B	ExpeL	100%	83.33	32.26	30.43	23.81	55.56	17.65	41.04
Llama-8B	ETO	0%	91.67	70.59	82.61	61.90	88.89	64.71	78.36
Llama-8B	KnowAgent	100%	87.50	93.55	65.22	66.67	61.11	64.71	75.37
Llama-8B	WKM	100%	95.83	87.10	86.96	61.90	66.67	52.94	77.61
Llama-8B	KnowSelf	15%	91.67	87.10	91.30	85.71	77.78	64.71	84.33
Gemma-2B	ReAct	0%	0.00	9.68	0.00	4.76	44.44	0.00	8.96
Gemma-2B	ETO	0%	91.67	83.87	78.26	52.38	77.78	29.41	71.64
Gemma-2B	KnowAgent	100%	91.67	90.32	69.57	71.43	66.67	41.18	73.88
Gemma-2B	WKM	100%	91.67	87.10	78.26	71.43	61.11	52.94	76.12
Gemma-2B	KnowSelf	26%	87.50	93.55	73.91	76.19	83.33	52.94	79.85

Success rates (%) on six ALFWorld task types. Highlighted rows: KnowSelf.

Minimal Knowledge, Maximum Impact

KnowSelf uses 15–26% knowledge queries vs. 100% for competing methods, yet achieves best scores



KnowSelf (Llama-8B): 84.33% accuracy with only 15% knowledge — surpasses GPT-4o ExpeL (79.85%) at 100% knowledge

Key Takeaways & Future Directions

- 1 Indiscriminate knowledge injection is wasteful — selective use consistently outperforms blanket application
- 2 Agents can learn situational self-awareness via special tokens with lightweight heuristic labeling
- 3 KnowSelf (Llama-8B) achieves 84.33%, surpassing GPT-4o ReAct (76.12%) and ExpeL (79.85%)
- 4 Only 15–26% knowledge usage needed versus 100% for KnowAgent, WKM, and ExpeL
- 5 Scales across model sizes — even Gemma-2B reaches 79.85%, matching GPT-4o ExpeL

Future Directions

- Extend to multi-modal agents and real-world robotics
- Explore dynamic knowledge base construction
- Apply self-awareness paradigm to multi-agent collaboration